

WRO Robot Wiring Map

1. ESP32 GPIO Map

Device	Signal	ESP32 GPIO
L298N	ENA	26
L298N	IN1	14
L298N	IN2	27
I2C	SDA	21
I2C	SCL	22
Servo	Signal	18
ToF Left	XSHUT	32
ToF Right	XSHUT	33
ToF Front	XSHUT	25
Color Sensors	S0	4
Color Sensors	S1	16
Color Sensors	S2	17
Color Sensors	S3	5
Ground Color	OUT	34
Front Color	OUT	35

2. I2C Expansion Board

Expansion Port	Connected To	Pins
Input	ESP32	3.3V, GND, SDA21, SCL22
Port 1	MPU6050	VCC GND SDA SCL
Port 2	ToF Left	VIN GND SDA SCL
Port 3	ToF Right	VIN GND SDA SCL
Port 4	ToF Front	VIN GND SDA SCL

3. Color Sensors

Pin	Ground Sensor	Front Sensor
VCC	5V	5V
GND	GND	GND
LED	5V	5V

S0	GPIO4	GPIO4
S1	GPIO16	GPIO16
S2	GPIO17	GPIO17
S3	GPIO5	GPIO5
OUT	GPIO34	GPIO35

4. Power

Part	Power	Notes
Servo	5V + GND	Signal GPIO18
L298N	Battery	Common GND
ESP32	USB / Shield	
ToF + MPU	3.3V via I2C board	
Color Sensors	5V	LED on 5V